

Suvi Häkkinen, Ph.D.

San Francisco Bay Area • (415) 676-0171 • shakkinen@berkeley.edu • smhakkinen.github.io

RESEARCH PROFILE

Neuroscientist with 10+ years of experience in human neuroimaging, neuroanatomy, and computational analysis. Current research bridges technical sequence optimization (VASO, GRASE) on NexGen 7T with mesoscale systems neuroscience to resolve laminar-specific brain organization.

EDUCATION

- Ph.D. in Psychology, University of Helsinki, Finland** 8/2018
Thesis: "Functional organization of human auditory cortex during active auditory tasks"
Supervisors: Prof. Teemu Rinne (University of Helsinki); Prof. C.G. Stecker (Vanderbilt University)
- M.Sc. (Tech.) in Bioinformation Technology, Aalto University, Finland** 10/2012
Major: Computational and Cognitive Biosciences
Minor: Acoustics and Audio Signal Processing
- B.Sc. (Tech.) in Bioinformation Technology, Aalto University, Finland** 8/2012
Major: Computational and Cognitive Biosciences
Minor: Usability in Software Development

RESEARCH EXPERIENCE

- Assistant Project Scientist** 9/2024 – Present
Postdoctoral Scholar 9/2023 – 9/2024

*Helen Wills Neuroscience Institute, University of California, Berkeley
Feinberg Lab (PI David Feinberg)*

- Scientific lead for a high-resolution hippocampal imaging project on the NexGen 7T, demonstrating that CBV-VASO provides superior depth-dependent localization by mitigating pial-vessel bias and venous contamination common in traditional BOLD-fMRI.
- Evaluated pulse sequence modifications and their effect on point-spread functions and activation patterns, providing empirical guidance for optimizing sequences for mesoscale imaging.

- Postdoctoral Scholar** 1/2022 – 9/2023

*Helen Wills Neuroscience Institute, University of California, Berkeley
Cognitive Neuroanatomy Lab (PI Kevin Weiner) & Building Blocks of Cognition Lab (PI Silvia Bunge)*

- Spearheaded a precision neuroanatomy project, investigating the functional relevance of cortical folding using a manually labeled dataset of 1,800+ sulci.
- Developed a multivariate analytical framework using SVM classification and graph analyses, demonstrating that individual sulcal morphology constitutes a promising "coordinate space".
- Mentored graduate students in surface-based fMRI analysis and statistics.

- Postdoctoral Fellow** 8/2018 – 12/2021
Memory and Aging Center, University of California, San Francisco (parental leave 5 mo)
Dementia Imaging Genetics Lab (PI Suzee Lee)

- Managed a project integrating data from 17 international sites to characterize neurodegenerative trajectories in presymptomatic and symptomatic *C9orf72*, *GRN*, and *MAPT* mutation carriers.
- Modeled lifespan trajectories of volumes, functional networks and fluid biomarkers using LME and normative-like modeling, identifying putative early biomarkers in the presymptomatic stage.
- Awarded a competitive personal research fellowship, securing two years of independent funding.

- Researcher** 1 – 7/2018

*Turku Brain and Mind Center, University of Turku, Finland
FinnBrain Birth Cohort Study (PI Hasse Karlsson)*

- Engineered robust fMRI preprocessing and connectivity pipelines optimized for neonatal imaging, supporting six peer-reviewed publications and four conference abstracts to date.

Graduate Researcher

Faculty of Medicine, University of Helsinki, Finland
@brain research group (PI Teemu Rinne)

9/2013 – 1/2018
(parental leave 10 mo)

- Headed five investigations into the functional architecture of human auditory cortex and its sensitivity to cognitive task requirements, with results published in three peer-reviewed journals.
- Established comprehensive analytical pipelines for fMRI, fMRS (choline-related plasticity), DWI (tractography), and EEG/MEG (source localization).
- Independently conducted 100+ 3T MRI sessions, assuming full responsibility for subject safety, protocol troubleshooting, and data acquisition.

Research Assistant

Institute of Psychology, University of Helsinki, Finland
Attention and Memory Networks research group (PIs Kimmo Alho and Teemu Rinne)

3/2008 – 12/2010
(part time)

- Operated 3T MRI systems (Siemens and GE platforms) to support data collection for audiovisual research studies.
- Developed a custom Python tool for generating flattened cortical patches, enabling 2D visualization of statistical maps in relation to morphology.

SKILLS

Neuroimaging

High-Resolution Expertise: Layer-fMRI • Hippocampal subfields • Auditory cortex • Cortical morphology

Analytical Frameworks: GLM/LME • Spatial nulls (spin testing) • Machine learning • Multivariate modeling • Graph theory

Diverse Populations: Neurodegenerative, neonatal and pediatric cohorts

Experimental Design: PsychoPy • Presentation

Technical Proficiencies

Programming: Python (NumPy, SciPy, pandas, Scikit-learn, TensorFlow) • MATLAB • R • C/C++ • Shell • SQL • LaTeX

Software: FreeSurfer • LayNII • Connectome Workbench • FSL • AFNI (including PTA for laminar modeling) • fMRIPrep • ANTs • Neuropointillist • NORDIC • ComBat

Computing & Infrastructure: HPC (SGE/Slurm) • Docker/Singularity • Git/GitHub • Nipype

Data standards: BIDS • GIFTI/CIFTI

Languages: Finnish (native) • English (full professional) • Swedish • German • French • Spanish

Specialized Training: Neural Networks, Optimization, and ML Strategy (Coursera/DeepLearning.AI, 2023) • Connectomics Summer School (Radboud University, 2018)

PUBLICATIONS (* denotes equal contribution/co-first authorship)

Peer-reviewed publications

Häkkinen, S. (née Talja), Voorhies, I.W., Willbrand, E.H., Yi-Heng, T., Gagnant, T., Yao, J.K., Weiner, K.S., Bunge, S.A. (2025) Anchoring functional connectivity to individual sulcal morphology yields insights in a pediatric study of reasoning. *Journal of Neuroscience*, 45(26), e0726242025.

Park, S., Beckett, A., **Häkkinen, S.**, Ma, S.J., Kim, H., Feinberg, D.A. (2025) Higher spatial resolution and sensitivity in whole brain functional MRI at 7T using 3D EPI accelerated with variable density CAIPI sampling and temporal random walk. *Magnetic Resonance in Medicine*, 94(2), 678–693.

Flagan, T. M., Chu, S.A., **Häkkinen, S.**, McFall, D., Heller, C., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. (2025) Functional connectivity associations with markers of disease progression in GRN mutation carriers. *Annals of Clinical and Translational Neurology*, 12(12), 2575–2588.

Vartiainen, E., Copeland, A. Pulli, E., Kumpulainen V., Silver, E. Rajasilta, O., Jolly, A., Luotonen, S., Kholis Audah, H., Bano, W., Suuronen, I., Saukko, E., **Häkkinen, S.**, Karlsson, H., Karlsson, L., Tuuluri, J.J. (2025) Pre -and postnatal maternal depressive symptoms associate with local connectivity of the left amygdala in 5-year-olds. *European Psychiatry*, 68(1), e130.

- Zhang, L., Flagan, T.M., **Häkkinen, S.**, Chu, S.A., Brown, J., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. (2023) Network connectivity alterations in presymptomatic and symptomatic MAPT mutation carriers. *Annals of Neurology*, 94(4), 632–646.
- Rajasilta, O., **Häkkinen, S.**, Björnsdotter, M., Scheinin, N. S., ... Tuulari, J.J. (2023) Maternal psychological distress associates with alterations in resting-state low-frequency fluctuations and distal functional connectivity of the neonate medial prefrontal cortex, *European Journal of Neuroscience*, 57, 242–257.
- Copeland, A., Korja, R., Nolvi, S., Pulli, E., Kumpulainen, V., Silver, E., Saukko, E., **Häkkinen, S.**, ... Tuulari, J.J. (2022) Maternal sensitivity at the age of 8 months associates with the child local synchrony of the medial prefrontal cortex at 5 years of age. *Frontiers in Neuroscience*, 16.
- Rajasilta, O., **Häkkinen, S.**, Björnsdotter, M., Scheinin, N. S., Lehtola, S. J., ... Tuulari, J.J. (2021) Increased maternal BMI during pregnancy is associated with altered neonate brain local synchrony in the left superior frontal gyrus, *Scientific Reports*, 11, 19182.
- Häkkinen, S.**, Chu, S.A., Lee, S.E. (2020) Neuroimaging in genetic frontotemporal dementia and amyotrophic lateral sclerosis. *Neurobiology of Disease*, 145, 205063.
- Rajasilta, O., Tuulari, J.J., Björnsdotter, M., Lehtola, S. J., Saunavaara, J., **Häkkinen, S.**, Merisaari, H., Parkkola, R., Lähdesmäki, T., Karlsson, L., Karlsson, H. (2020) Resting state networks of the neonate brain identified in independent component analysis. *Developmental Neurobiology*, 80, 111–125.
- Häkkinen, S.**, Rinne, T. (2018) Intrinsic, stimulus-driven and task-dependent connectivity in human auditory cortex. *Brain Structure & Function*, 223(5), 2113–2127.
- Häkkinen, S.**, Ovaska, N., Rinne, T. (2015) Processing of pitch and location in human auditory cortex during visual and auditory tasks. *Frontiers in Psychology*, 6.
- Talja, S.**, Alho, K., Rinne, T. (2015) Source analysis of event-related potentials during pitch discrimination and pitch memory tasks. *Brain Topography*, 28, 445–458.
- Rinne, T., Koistinen, S., **Talja, S.**, Wikman, P., Salonen, O. (2012) Task-dependent activations of human auditory cortex during spatial discrimination and spatial memory tasks. *NeuroImage*, 59, 4126–4131.

Works currently under review

- Häkkinen, S.**, Beckett, A., Walker, E., Huber, L., Feinberg, D.A. Functional imaging of hippocampal layers using VASO on the Next Generation (NexGen) 7T, <https://doi.org/10.1101/2025.08.29.673151>
- Zhang, L.*, **Häkkinen, S.***, Chu, S. A., Flagan, T. M., Coppola, G., Geschwind, D. H., ... Lee, S. E., on behalf of the ARTFL/LEFFTDS/ALLFTD Consortia. Longitudinal functional network connectivity changes across the clinical stages of C9orf72 hexanucleotide repeat expansion carriers, <https://doi.org/10.1101/2025.11.23.25340528>
- Copeland, A., Rajasilta, O., **Häkkinen, S.**, ..., Karlsson, H., Tuulari, J.J. Functional network organization of the 5-year-old brain identified using independent component analysis and evaluation of denoising strategies, submitted.

CONFERENCE ABSTRACTS

- Ru, H., **Häkkinen, S.**, Beckett, A., Walker, E., Feinberg, D.A., Park, S. Accelerated 3D Zoomed GRASE for Mesoscale fMRI using Self-Supervised Reconstruction (DeepGRASE) [Abstract submitted for presentation]. *OHBM*, 2026.
- Beckett, A., Walker, E.B., **Häkkinen, S.**, Khagai, O., Vu, A., Huber, R., Feinberg, D. Studying cortical networks using whole brain cerebral blood volume weighted imaging on the NexGen 7T [Abstract submitted for presentation]. *OHBM*, 2026.
- Beckett, A.J., **Häkkinen, S.**, Walker, E.B., Khagai, O., Vu, A., Huber, R., & Feinberg, D. Whole-brain, cerebral blood volume weighted imaging of cortical networks using the Next Generation (NexGen) 7T scanner [Abstract submitted for presentation]. *ISMRM*, 2026.
- Häkkinen, S.**, Voorhies, W.I., Willbrand, E.H., Tsai, Y.-H., Yao, J.K., Weiner, K.S., Bunge, S.A. Lateral frontoparietal functional connectivity based on individual sulcal morphology. *SfN*, 2025.
- Häkkinen, S.**, Beckett, A., Walker, E., Huber, R., Feinberg, D.A. Functional imaging of hippocampal layers using VASO on the Next Generation (NexGen) 7T. *ISMRM*, 2025.

- Beckett, A.*, **Häkkinen, S.***, Walker, E., Huber, R., Feinberg, D.A. Functional imaging of hippocampal layers using VASO on the Next Generation (NexGen) 7T. *ISMRM Joint Workshop of the Ultra-High Field MR & Brain Function Study Groups*, 2025.
- Walker, E. B, Beckett, A. J., **Häkkinen, S.**, Vu, A., Huber, R., & Feinberg, D. Whole-brain, cerebral blood volume weighted imaging of cortical networks using the Next Generation (NexGen) 7T scanner. *SfN*, 2025
- Beckett, A., Park, S., **Häkkinen, S.**, Walker, E., Vu, A., Feinberg, D.A. Development of GRASE Pulse Sequence with larger field of view for mesoscale functional MRI on the Next Generation (NexGen) 7T scanner. *ISMRM*, 2025.
- Zhang, L., **Häkkinen, S.**, Jung, Y., Mandelli, M.L., Leichter, D., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS/ALLFTD Consortia.. *AAIC*, 2025. Longitudinal functional connectivity changes across the clinical spectrum of *C9orf72* hexanucleotide repeat expansion carriers. *AAIC*, 2025.
- Ma, S.J., Walker, E., **Häkkinen, S.**, Beckett, A.J.S., Vu, A.T., Feinberg, D.A. NexGen 7T MRI scanner for mesoscale brain imaging. *ANA*, 2024.
- Flagan, T.M., Chu, S.A., **Häkkinen, S.**, Zhang, L., ... Seeley, W.W., Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. Functional connectivity associations with markers of disease progression in *GRN* mutation carriers. *AAIC*, 2024.
- Huber, R., Beckett, A., Ma, S., Stirnberg, R., Morgan, T., Ehses, P., **Häkkinen, S.**, Gunamony, S., Bandettini, P., Feinberg, S. Developing whole brain layer-fMRI protocols on the NexGen 7T scanner. *OHBM*, 2023.
- Beckett, A.J.S., Huber, R., Ma, S. J., **Häkkinen, S.**, Gunamony, S., Feinberg, D.A. Whole brain layer-fMRI on the NexGen 7T scanner with high performance gradients and 64-channel receiver array. *ISMRM*, 2023.
- Park, S., Beckett, A.J.S., **Häkkinen, S.**, Ma, S. J., Feinberg, D.A. Whole-brain sub-millimeter resolution fMRI with 3D EPI accelerated with temporal random walk. *ISMRM*, 2023.
- Lombardi, J., Zhang, L., Flagan, T.M., Vargas, A.G., Mandelli, M.L., Brown, J.A., **Häkkinen, S.**, ... Miller, B.L., Gorno-Tempini, M.L., Seeley, W.W., Lee, S.E. on behalf of the ALLFTD Consortium. Structural and functional alterations in young adult presymptomatic frontotemporal lobar degeneration mutation carriers. *AAIC*, 2023.
- Zhang, L., Flagan, T.M., Staffaroni, A., **Häkkinen, S.**, Brown, J.A., Mandelli, M.L., ... Gorno-Tempini, M.L., Miller, B.L., Seeley, W.W., Lee, S.E. on behalf of the ARTFL/LEFFTDS/ALLFTD Consortia. Functional connectivity trajectories in genetic frontotemporal dementia. *AAIC*, 2023.
- Lombardi, J., Zhang, L., Flagan, T. M., Mandelli, M. L., Brown, J. A., **Häkkinen, S.**, ... Lee, S.E., on behalf of the ALLFTD Consortium. Salience network alterations in young adult presymptomatic FTLD mutation carriers. *ISFTD*, 2022.
- Copeland, A., Korja, R., Nolvi, S., Pulli, E., Kumpulainen, V., Silver, E., Saukko, E., **Häkkinen, S.**, ... Tuulari, J.J. Maternal sensitivity at the age of 8 months associates with the child local synchrony of the medial prefrontal cortex at 5 years of age. *Flux Congress 2022*.
- Flagan T. M., Chu S.A., **Häkkinen S.**, McFall, D., Heller, C., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. Complement and NfL associations with brain structure and functional connectivity alterations in presymptomatic and symptomatic *GRN* mutation carriers. *AAIC 2021*.
- Zhang, L., Flagan, T.M., Chu, S.A., **Häkkinen, S.**, Brown, J., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. Presymptomatic and symptomatic *MAPT* mutation carriers feature functional connectivity alterations. *AAIC*, 2021.
- Häkkinen, S.**, Chu, S.A., Flagan, T. M., Gendron, T. D., Petrucelli, L., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. Associations of poly(GP), NfL and functional network alterations in *C9orf72* expansion carriers. *OHBM*, 2020.
- Zhang, L., Flagan, T. M., Chu, S.A., **Häkkinen, S.**, Rojas-Martinez, J., ... Lee, S.E., on behalf of the ALLFTD Consortia. Brain functional connectivity changes in presymptomatic and symptomatic *MAPT* mutation carriers. *OHBM*, 2020.
- Tuulari, J.J., **Häkkinen, S.**, Rajasilta, O., Lehtola, S., Merisaari, H., ... Karlsson, H. Maternal prenatal anxiety associates to infant orbitofrontal cortex functional synchrony. *Flux*, 2018.
- Rajasilta, O., Tuulari J.J., **Häkkinen, S.**, Lehtola, S., Merisaari, H., ... Karlsson, H. Functional networks in term-born infants – a resting-state functional MRI study from FinnBrain Birth Cohort study. *Flux*, 2018.

Rinne, T., **Talja, S.**, Stecker, G.C. (2015) Processing in auditory cortex depends on the characteristics of the discrimination task: fMRI study. *ARO MWM*, 2015.

Stecker, G.C., Higgins, N., McLaughlin, S., **Talja, S.**, Rinne, T. (2015) Stimulus- and task-dependent spatial processing in the human auditory cortex. *ARO MWM*, 2015.

Talja, S., Ovaska, N., Rinne, T. (2014) Feature-specific and task-dependent processing of pitch and location in human auditory cortex. *OHBM*, 2014.

Rinne, T., Ovaska, N., **Talja, S.** (2013) Task-dependent activations of human auditory cortex during pitch and spatial tasks. *ARO MWM*, 2013.

DATASETS

Häkkinen, S., Beckett, A., Walker, E., Huber, L., Feinberg, D. (2025). 7T fMRI VASO of the human hippocampus. OpenNeuro. [Dataset] doi: doi:10.18112/openneuro.ds007122.v1.0.0

GRANTS & FELLOWSHIPS

Päivi and Sakari Sohlberg Foundation 140,000 € for postdoctoral research investigating genetic dementia biomarkers	2019
Instrumentarium Science Foundation 18,000 € for doctoral research on auditory neuroscience	2016
University of Helsinki Doctoral Fellowship (3-year full funding) Selected by the Doctoral Programme of Psychology, Learning, and Communication.	2014

MENTORSHIP & EDUCATIONAL OUTREACH

Research Mentor, Feinberg, Weiner and Bunge Labs, UC Berkeley • Guided graduate students in MRI analysis, experimental design, and scientific writing.	2022 – Present
Discussion Lead, Cognitive Neuroscience Colloquium, UC Berkeley • "Brain-wide association studies: Challenges and opportunities"	2022
Speaker, MAC Postdoc Educational Curriculum, Memory and Aging Center, UCSF • "NMR basics and MRI contrasts"	2019
Research Mentor, @brain research group, University of Helsinki • Advised a student through analyses and writing of a Master's thesis on auditory neuroscience. • Tutored interns in subject recruitment and behavioral training.	2011 – 2013

PROFESSIONAL RECOGNITION & SERVICE

Invited Speaker, layer-fMRI talk (webinar for NIH/Maastricht/MGH researchers), 2026

Best Poster Award (Joint 1st Author): ISMRM Joint Workshop of the Ultra-High Field MR & Brain Function Study Groups, 2025.

Ad-Hoc Reviewer: *Human Brain Mapping, Developmental Cognitive Neuroscience*

Board Member (webmaster 2012 – 2016, secretary 2013 – 2014), Division of Young Researchers, Finnish Psychological Society

Founding Board Member (secretary 2004 – 2006), Bioinformation Technology Student Association, Helsinki University of Technology

Public Outreach: Research on tertiary sulci featured in *UC Berkeley News*

PROFESSIONAL MEMBERSHIPS

Society for Neuroscience (SfN)	2025
The International Society for Magnetic Resonance in Medicine (ISMRM)	2025
Organization for Human Brain Mapping (OHBM)	2014, 2020